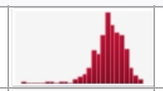
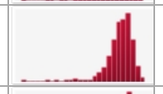
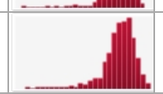
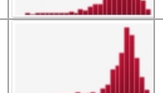
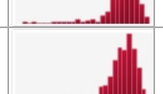
@RISK - Results Summary

Report: Outputs
Performed By: mehak
Date: 01 July 2026

Name	Simulation	Cell	Function	Graph	Minimum	Maximum	Mean	Mode	Median	Std. Deviation	1%	99%	Errors	Filtered
Range: <none>														
Queue>10	Sim 1	N78	RiskOutput("Queue>10")		0	1	0.1740	0	0	0.3795	0	1	0	0
Queue>10	Sim 2	N78	RiskOutput("Queue>10")		0	1	0.3800	0	0	0.4859	0	1	0	0
Queue>10	Sim 3	N78	RiskOutput("Queue>10")		0	1	0.4620	0	0	0.4991	0	1	0	0
Queue>10	Sim 4	N78	RiskOutput("Queue>10")		0	1	0.5740	1	1	0.4950	0	1	0	0
Queue>10	Sim 5	N78	RiskOutput("Queue>10")		0	1	0.6860	1	1	0.4646	0	1	0	0
Range: Total Profit														
Total Profit / Profit	Sim 1	N76	RiskOutput()		-18,780	2,680	-3,198.08	-2,840	-3,120	2,536.77	-10,680	1,800	0	0
Total Profit / Profit	Sim 2	N76	RiskOutput()		-17,860	26,120	17,328.16	15,380	18,020	4,598.38	-160	24,480	0	0
Total Profit / Profit	Sim 3	N76	RiskOutput()		-6,720	33,360	23,446.96	24,040	24,200	5,271.39	1,440	31,800	0	0
Total Profit / Profit	Sim 4	N76	RiskOutput()		-26,460	36,760	24,211.40	26,420	25,640	7,475.48	-5,800	35,280	0	0
Total Profit / Profit	Sim 5	N76	RiskOutput()		-23,860	37,560	22,606.56	27,300	24,280	8,570.57	-13,340	34,460	0	0

Based on this result, Scenario 4 that is replacing the heating element after 4 days of use; provides the highest average profit of nearly $24,211 over the 60 day period. In addidition to this, with this policy, the estimated probability that end of day queue exceeds 10 castings atleast once during this period is approximately 57.4%.

Therefore, I recommend using a 4-day preventive replacement policy, as it provides highest expected profit as compared to other alternatives available.